

# Table-leg zip-tie drawer set

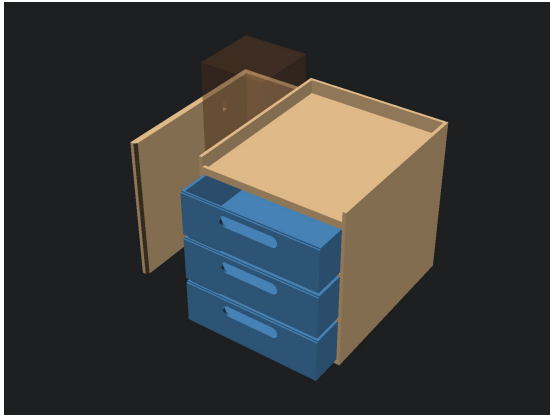
Ten printable variants of one OpenSCAD file. Housing and drawers are separate parts. Housing always prints with drawer openings up (back of the housing on the bed). Drawers always print bottom-down. A C-channel on the left or right takes a table leg and zip-ties through windows in the outer wall. Choose a winner from the plates below.

## Print envelope

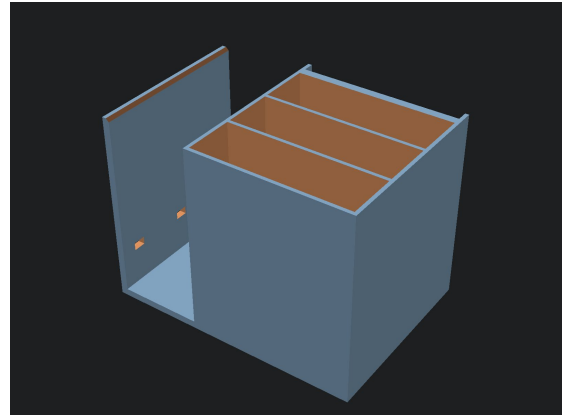
Target printer: Bambu Lab H2D. Single-nozzle volume is 325 x 320 x 325 mm. This model uses a 5 mm guard of **320 x 315 x 320 mm**. Every variant in this catalog is marked PASS against that guard. File: `leg-drawers.scad` plus `variants.json`.

## How the parts go together

The housing is a stack of open-front bays with a tray on top. Tray wall heights are independent: left, right, back, and a usually-lower front lip. The C-channel is open at the front so you can slide it onto a standing table leg, then zip-tie through the outer-wall windows. Drawers are separate open-top boxes with a finger slot in the front wall.



*Assembled (office-slim-left)*



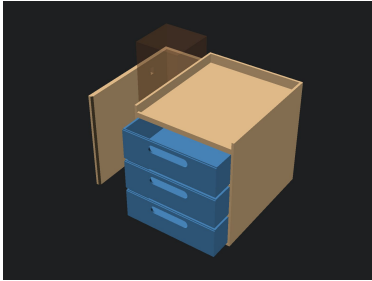
*Housing print pose — openings up*

## Comparison

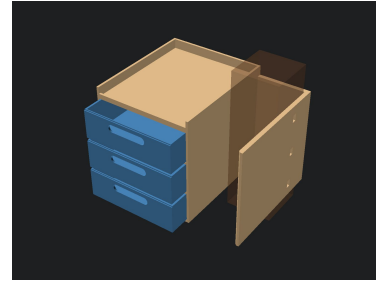
| Name              | Drawers | Bay H x W x D mm | Leg X x Y mm | Mount | Ledges L/R/B/F mm | Housing print mm | H2D  |
|-------------------|---------|------------------|--------------|-------|-------------------|------------------|------|
| office-slim-left  | 3       | 32 x 100 x 120   | 50 x 50      | left  | 10/10/10/3        | 161 x 116 x 122  | PASS |
| office-slim-right | 3       | 32 x 100 x 120   | 50 x 50      | right | 10/10/10/3        | 161 x 116 x 122  | PASS |
| pencil-four       | 4       | 20 x 85 x 130    | 50 x 50      | left  | 8/8/8/2           | 146 x 100 x 132  | PASS |
| tool-two-deep     | 2       | 55 x 130 x 170   | 75 x 75      | left  | 14/14/14/5        | 216 x 131 x 172  | PASS |
| ikea-adils        | 3       | 36 x 110 x 125   | 40 x 40      | left  | 10/10/10/3        | 161 x 128 x 127  | PASS |
| rect-rail-leg     | 3       | 38 x 140 x 120   | 40 x 80      | left  | 12/12/18/3        | 191 x 142 x 122  | PASS |
| tray-top          | 3       | 40 x 120 x 140   | 50 x 50      | left  | 28/28/28/8        | 181 x 158 x 142  | PASS |
| mini-right        | 2       | 28 x 75 x 85     | 35 x 35      | right | 6/6/6/2           | 121 x 69 x 87    | PASS |
| wide-three        | 3       | 42 x 180 x 160   | 70 x 70      | left  | 12/12/12/4        | 261 x 148 x 162  | PASS |
| asymmetric-wall   | 3       | 36 x 115 x 125   | 60 x 40      | right | 22/6/22/2         | 186 x 140 x 127  | PASS |

Housing print axes after the openings-up rotation: X = width + mount, Y = stacked height + tallest ledge, Z = housing depth (front-back). Drawer print axes: X = width, Y = depth, Z = height.

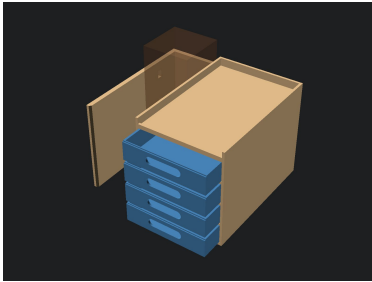
## Assembled ISO — all ten



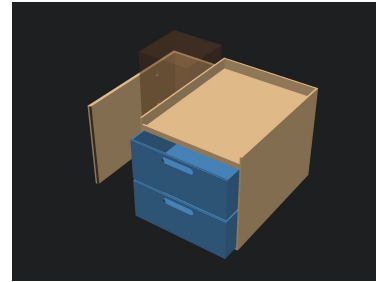
*office-slim-left*



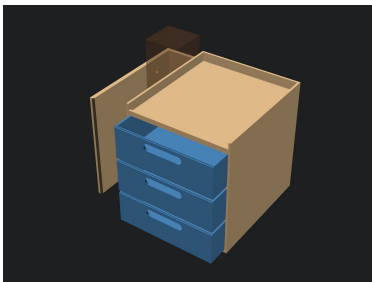
*office-slim-right*



*pencil-four*



*tool-two-deep*



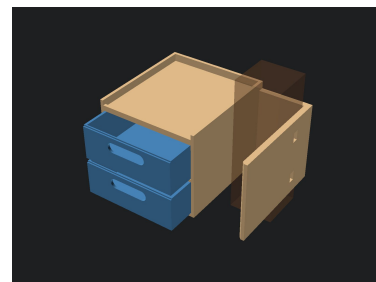
*ikea-adils*



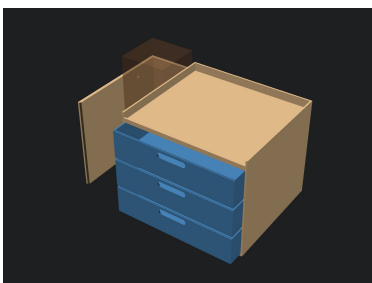
*rect-rail-leg*



*tray-top*



*mini-right*



*wide-three*

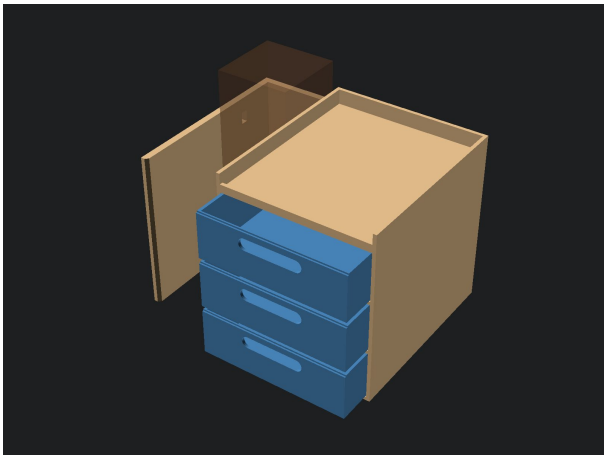


*asymmetric-wall*

# Office slim, left mount (office-slim-left)

Everyday desk: three shallow drawers on a 50 mm square leg, mount on the left.

| Parameter                 | Value | Parameter          | Value               |
|---------------------------|-------|--------------------|---------------------|
| drawer_count              | 3     | leg_on_left        | left                |
| drawer_height mm          | 32    | leg_x mm           | 50                  |
| drawer_width mm           | 100   | leg_y mm           | 50                  |
| drawer_depth mm           | 120   | zip ties (auto)    | 3                   |
| ledge_left mm             | 10    | housing print X mm | 160.9               |
| ledge_right mm            | 10    | housing print Y mm | 115.6               |
| ledge_back mm             | 10    | housing print Z mm | 122.4               |
| ledge_front mm            | 3     | drawer print mm    | 99.1 x 119.2 x 31.6 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 99.1 x 119.2 x 31.6 |



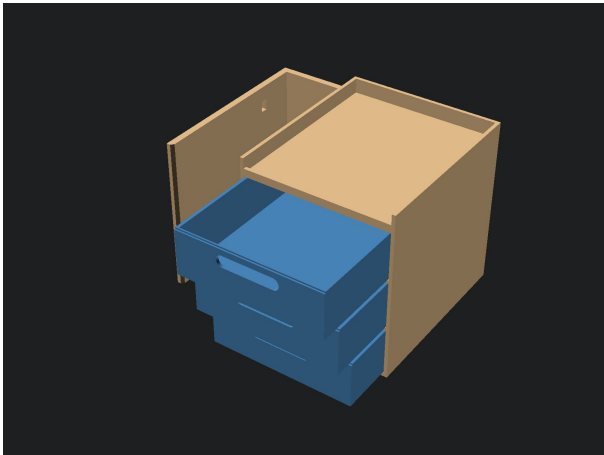
ISO, drawers pulled, ghost leg



ISO from the rear / mount

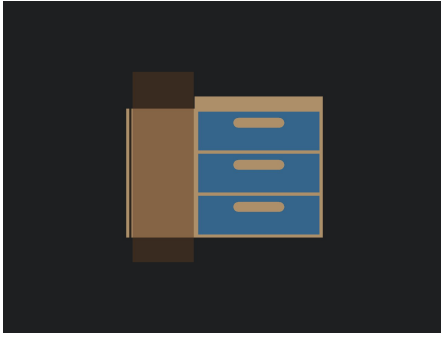


ISO opposite flank



Exploded — housing and drawers separate

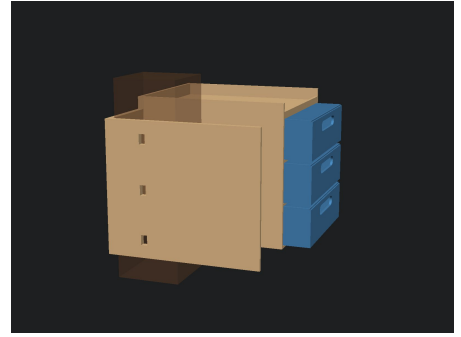
## Office slim, left mount — drawings and print poses



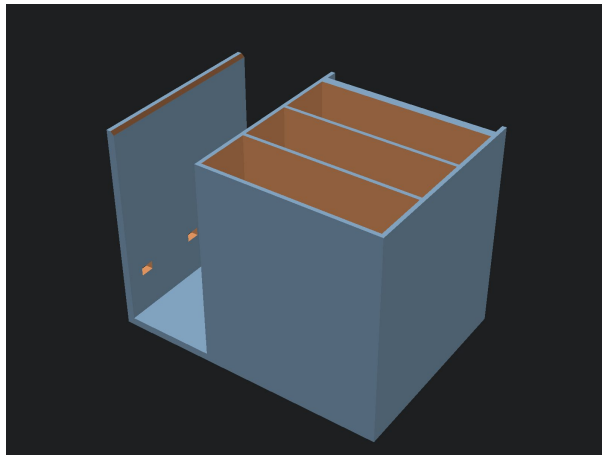
*Front elevation*



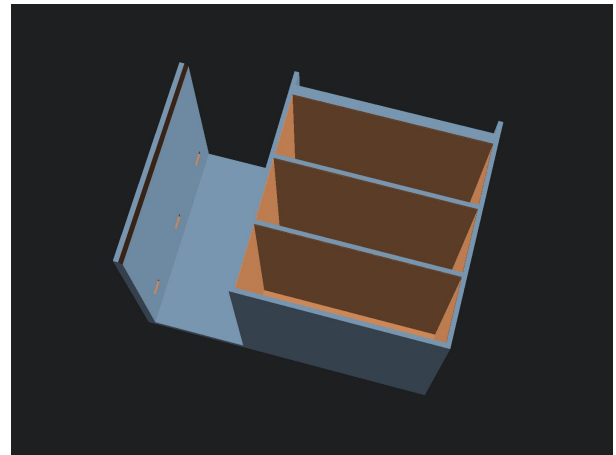
*Top / plan (C-channel)*



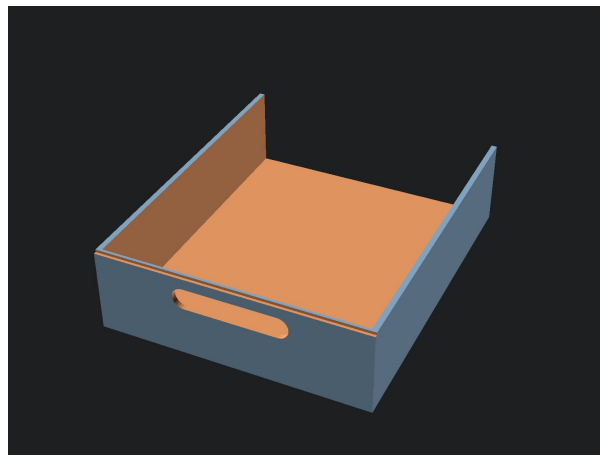
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



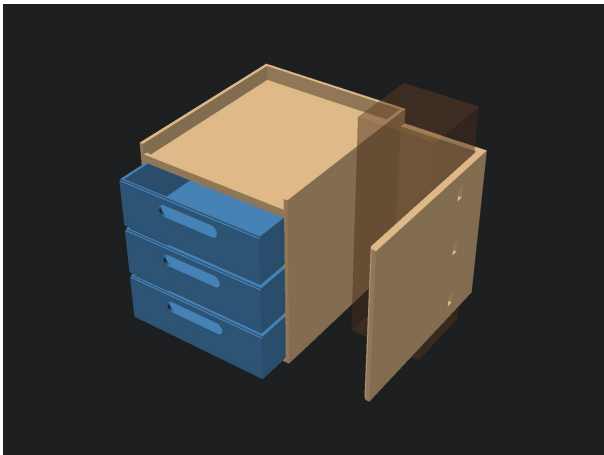
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/office-slim-left/housing_print.stl` and `export/office-slim-left/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

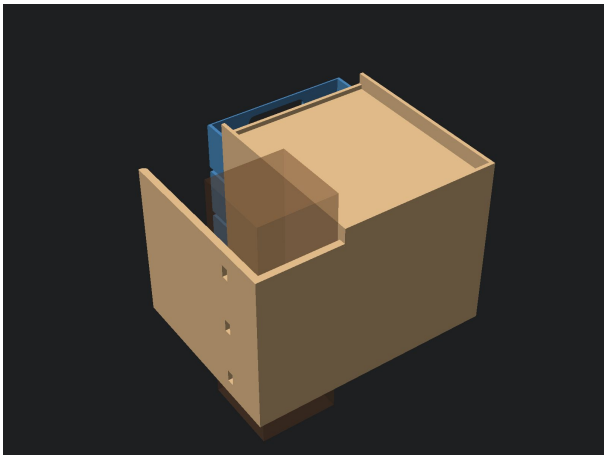
# Office slim, right mount (office-slim-right)

Mirror of office-slim-left for the right-hand table leg.

| Parameter                 | Value | Parameter          | Value               |
|---------------------------|-------|--------------------|---------------------|
| drawer_count              | 3     | leg_on_left        | right               |
| drawer_height mm          | 32    | leg_x mm           | 50                  |
| drawer_width mm           | 100   | leg_y mm           | 50                  |
| drawer_depth mm           | 120   | zip ties (auto)    | 3                   |
| ledge_left mm             | 10    | housing print X mm | 160.9               |
| ledge_right mm            | 10    | housing print Y mm | 115.6               |
| ledge_back mm             | 10    | housing print Z mm | 122.4               |
| ledge_front mm            | 3     | drawer print mm    | 99.1 x 119.2 x 31.6 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 99.1 x 119.2 x 31.6 |



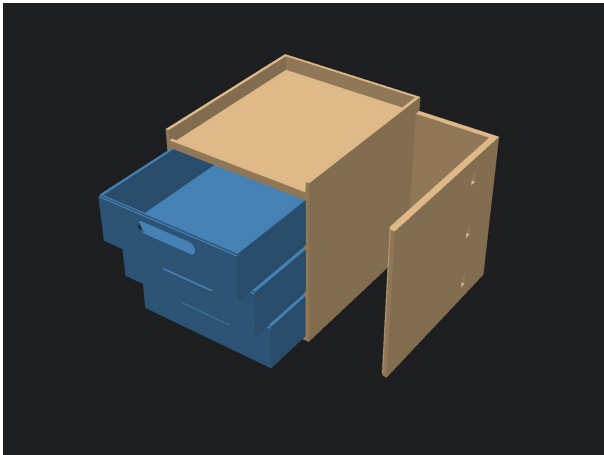
ISO, drawers pulled, ghost leg



ISO from the rear / mount



ISO opposite flank



Exploded — housing and drawers separate

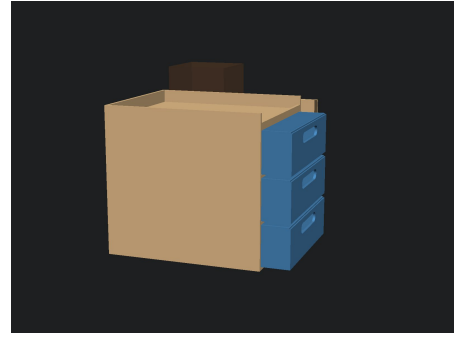
## Office slim, right mount — drawings and print poses



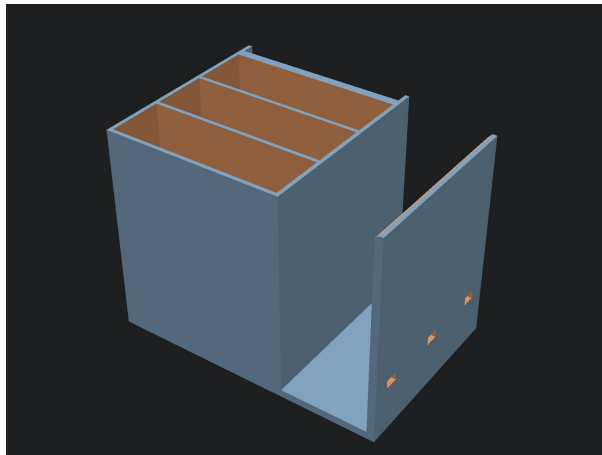
*Front elevation*



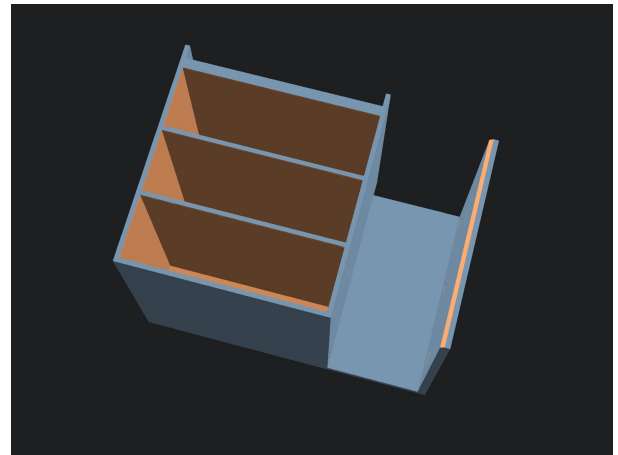
*Top / plan (C-channel)*



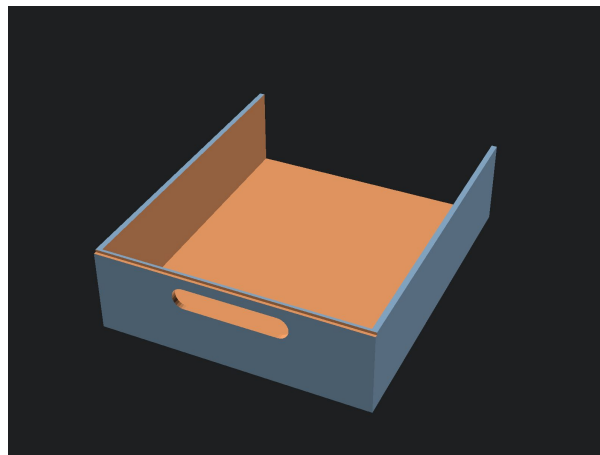
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



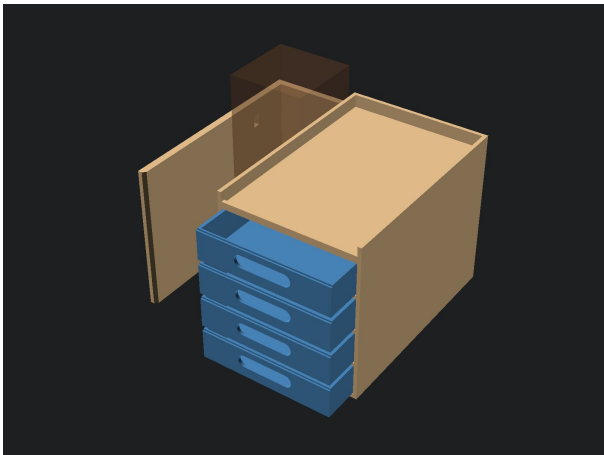
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/office-slim-right/housing_print.stl` and `export/office-slim-right/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# Pencil four-stack (pencil-four)

Four low bays for pens, bits, and SD cards. 50 mm square leg, left mount.

| Parameter                 | Value | Parameter          | Value               |
|---------------------------|-------|--------------------|---------------------|
| drawer_count              | 4     | leg_on_left        | left                |
| drawer_height mm          | 20    | leg_x mm           | 50                  |
| drawer_width mm           | 85    | leg_y mm           | 50                  |
| drawer_depth mm           | 130   | zip ties (auto)    | 3                   |
| ledge_left mm             | 8     | housing print X mm | 145.9               |
| ledge_right mm            | 8     | housing print Y mm | 100.0               |
| ledge_back mm             | 8     | housing print Z mm | 132.4               |
| ledge_front mm            | 2     | drawer print mm    | 84.1 x 129.2 x 19.6 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 84.1 x 129.2 x 19.6 |



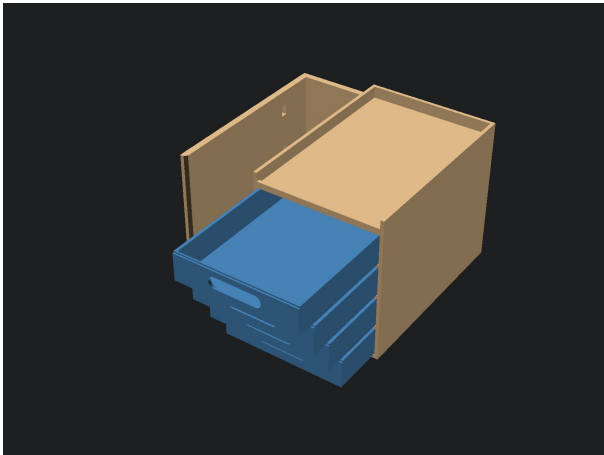
ISO, drawers pulled, ghost leg



ISO from the rear / mount

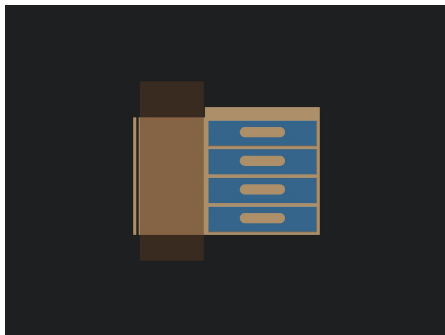


ISO opposite flank

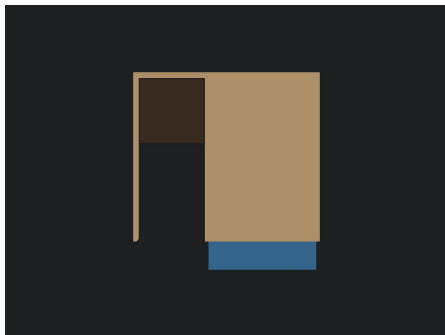


Exploded — housing and drawers separate

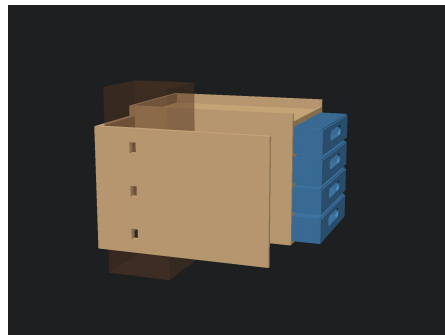
## Pencil four-stack — drawings and print poses



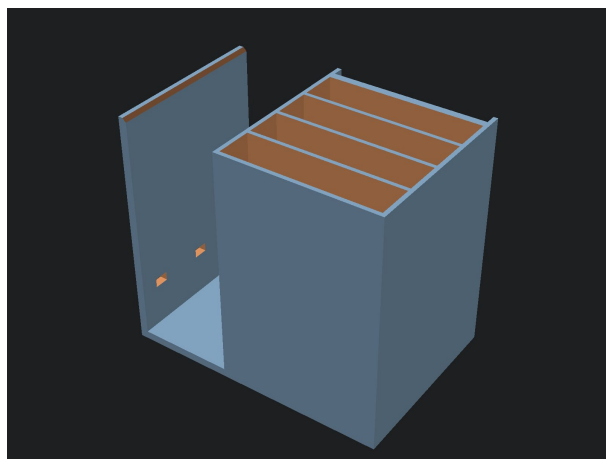
*Front elevation*



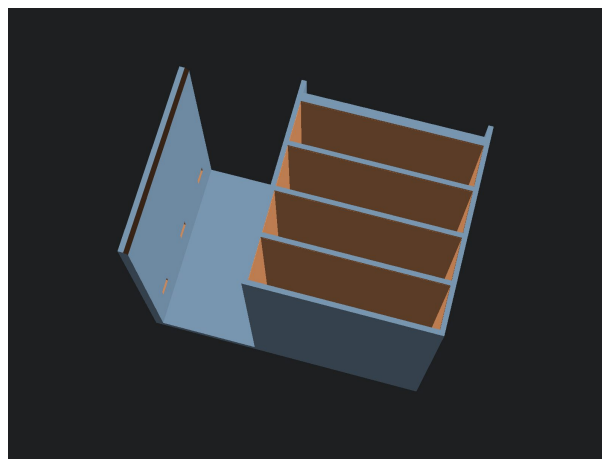
*Top / plan (C-channel)*



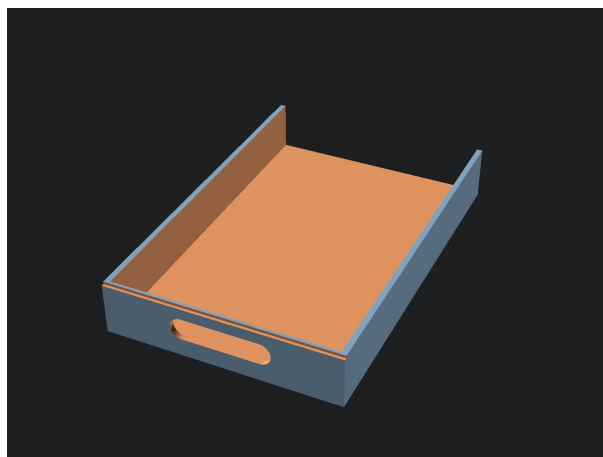
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



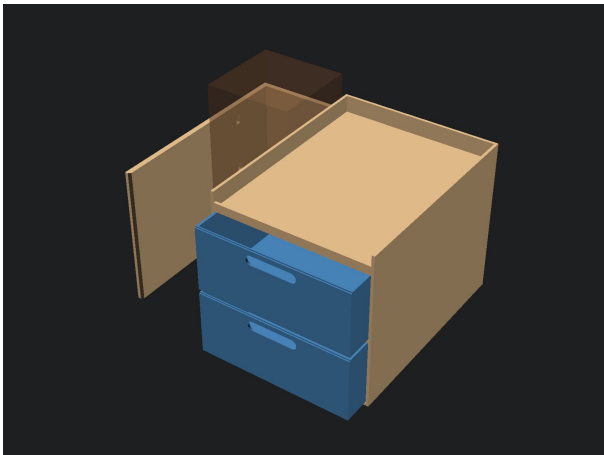
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/pencil-four/housing_print.stl` and `export/pencil-four/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 4 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# Tool two-deep (tool-two-deep)

Two tall bays for hand tools. 75 mm square leg, left mount, taller tray walls.

| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 2     | leg_on_left        | left                 |
| drawer_height mm          | 55    | leg_x mm           | 75                   |
| drawer_width mm           | 130   | leg_y mm           | 75                   |
| drawer_depth mm           | 170   | zip ties (auto)    | 3                    |
| ledge_left mm             | 14    | housing print X mm | 215.9                |
| ledge_right mm            | 14    | housing print Y mm | 131.2                |
| ledge_back mm             | 14    | housing print Z mm | 172.4                |
| ledge_front mm            | 5     | drawer print mm    | 129.1 x 169.2 x 54.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 129.1 x 169.2 x 54.5 |



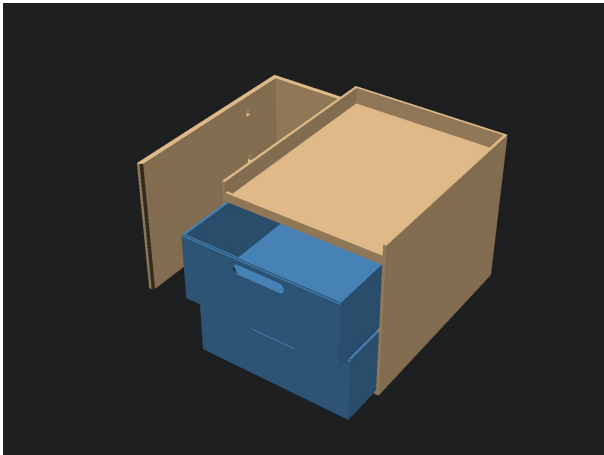
ISO, drawers pulled, ghost leg



ISO from the rear / mount

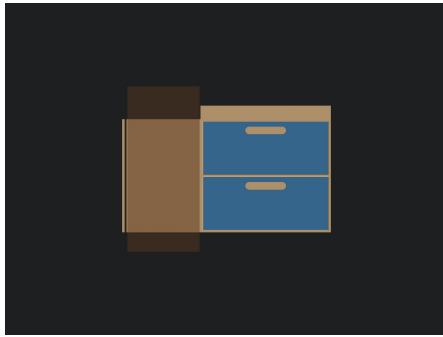


ISO opposite flank

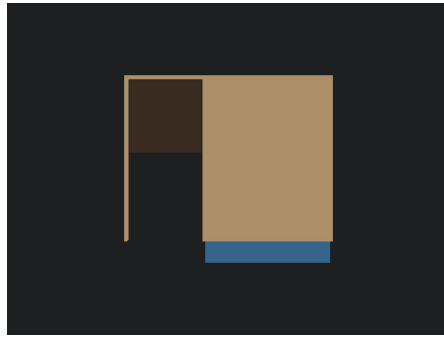


Exploded — housing and drawers separate

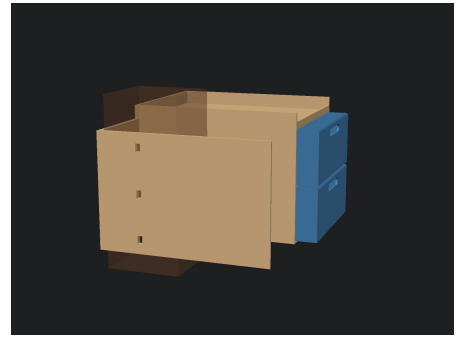
## Tool two-deep — drawings and print poses



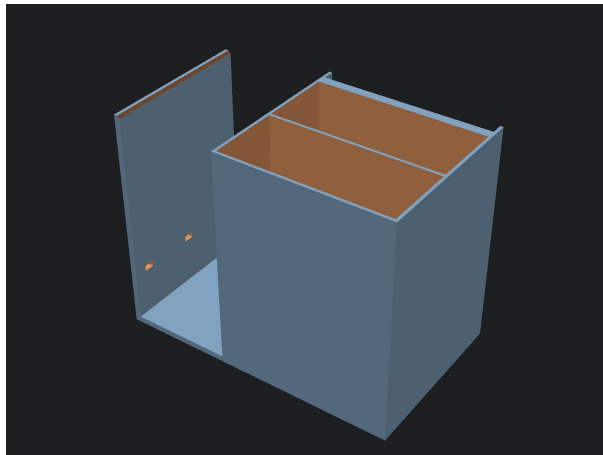
*Front elevation*



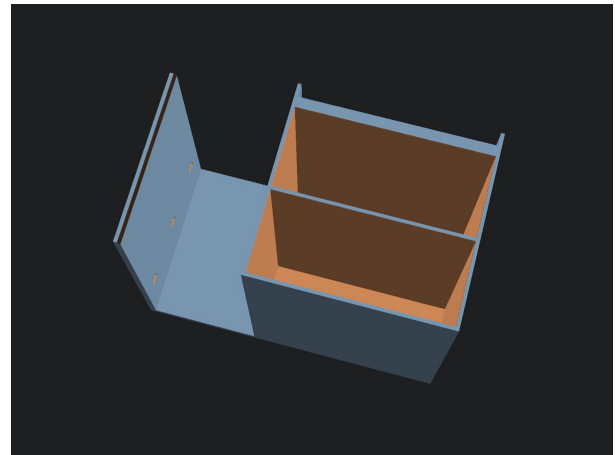
*Top / plan (C-channel)*



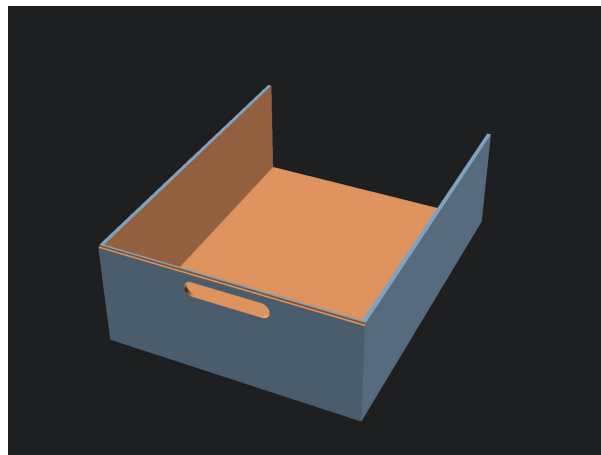
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



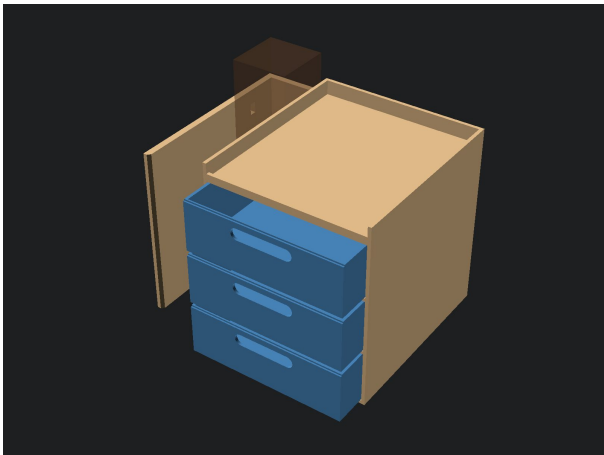
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/tool-two-deep/housing_print.stl` and `export/tool-two-deep/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 2 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# IKEA ADILS-size pocket (ikea-adils)

40 mm square pocket (covers ~38 mm ADILS-class round legs with clearance), three mid bays.

| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 3     | leg_on_left        | left                 |
| drawer_height mm          | 36    | leg_x mm           | 40                   |
| drawer_width mm           | 110   | leg_y mm           | 40                   |
| drawer_depth mm           | 125   | zip ties (auto)    | 3                    |
| ledge_left mm             | 10    | housing print X mm | 160.9                |
| ledge_right mm            | 10    | housing print Y mm | 127.6                |
| ledge_back mm             | 10    | housing print Z mm | 127.4                |
| ledge_front mm            | 3     | drawer print mm    | 109.1 x 124.2 x 35.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 109.1 x 124.2 x 35.5 |



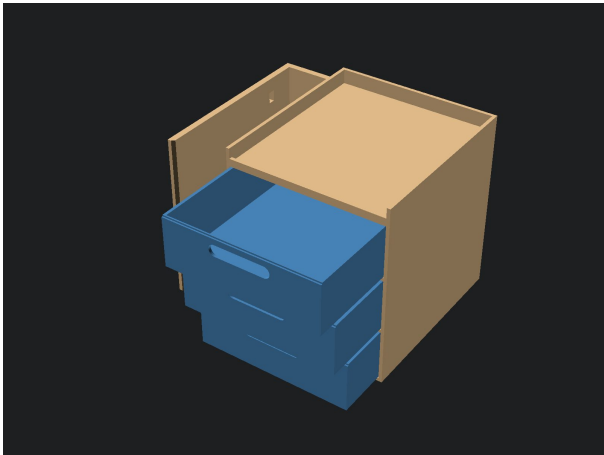
ISO, drawers pulled, ghost leg



ISO from the rear / mount



ISO opposite flank



Exploded — housing and drawers separate

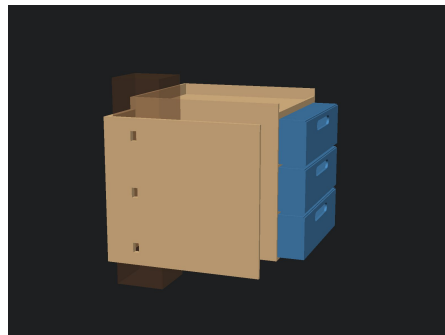
## IKEA ADILS-size pocket — drawings and print poses



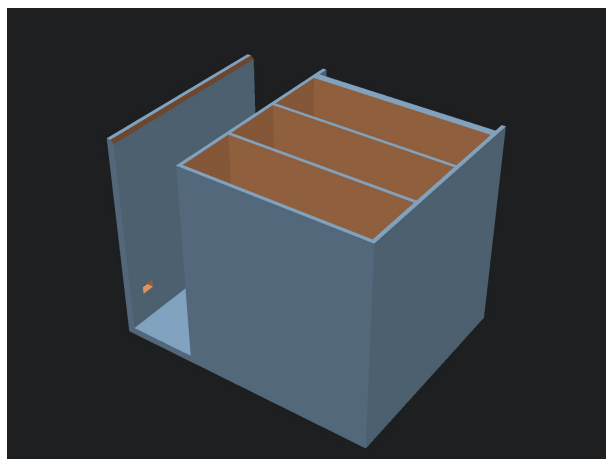
*Front elevation*



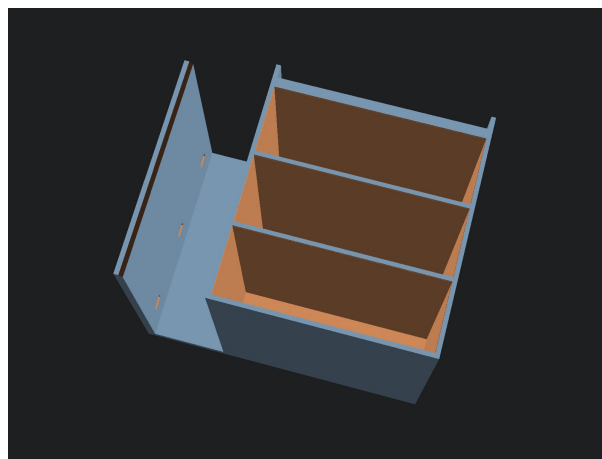
*Top / plan (C-channel)*



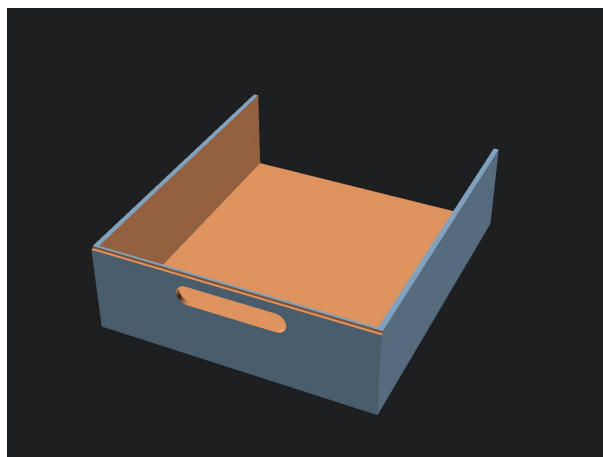
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



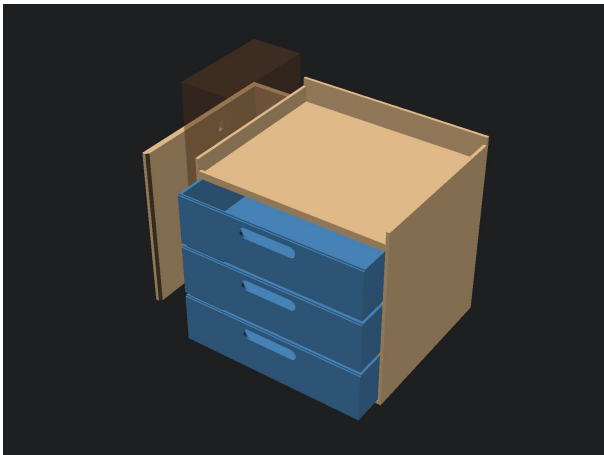
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/ikea-adils/housing_print.stl` and `export/ikea-adils/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# Rectangular rail leg (rect-rail-leg)

40 x 80 mm rectangular desk rail. Tall back ledge to sit against the apron.

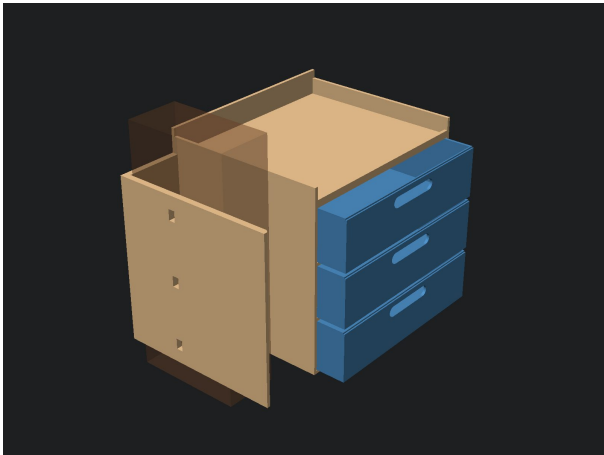
| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 3     | leg_on_left        | left                 |
| drawer_height mm          | 38    | leg_x mm           | 40                   |
| drawer_width mm           | 140   | leg_y mm           | 80                   |
| drawer_depth mm           | 120   | zip ties (auto)    | 3                    |
| ledge_left mm             | 12    | housing print X mm | 190.9                |
| ledge_right mm            | 12    | housing print Y mm | 141.6                |
| ledge_back mm             | 18    | housing print Z mm | 122.4                |
| ledge_front mm            | 3     | drawer print mm    | 139.1 x 119.2 x 37.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 139.1 x 119.2 x 37.5 |



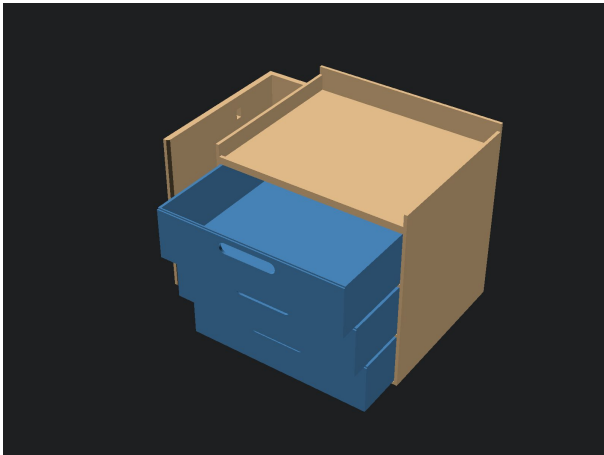
ISO, drawers pulled, ghost leg



ISO from the rear / mount



ISO opposite flank



Exploded — housing and drawers separate

## Rectangular rail leg — drawings and print poses



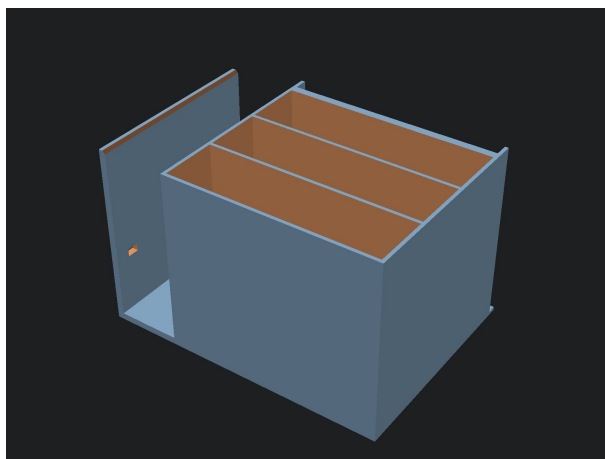
*Front elevation*



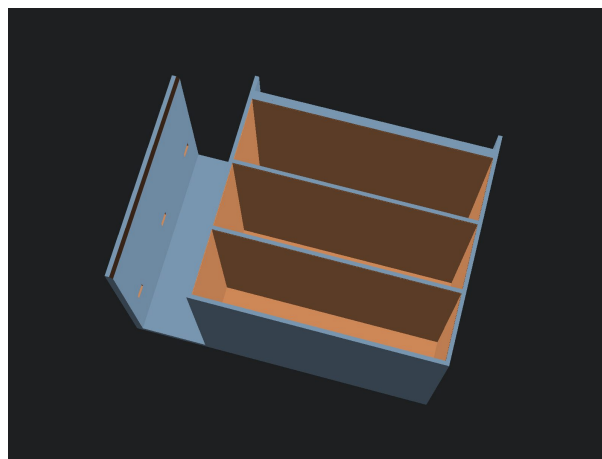
*Top / plan (C-channel)*



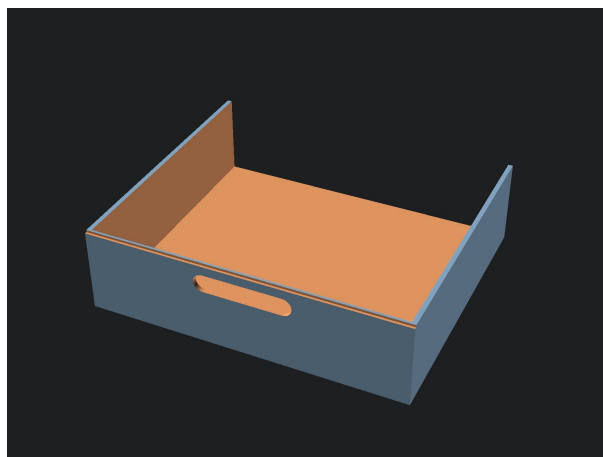
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



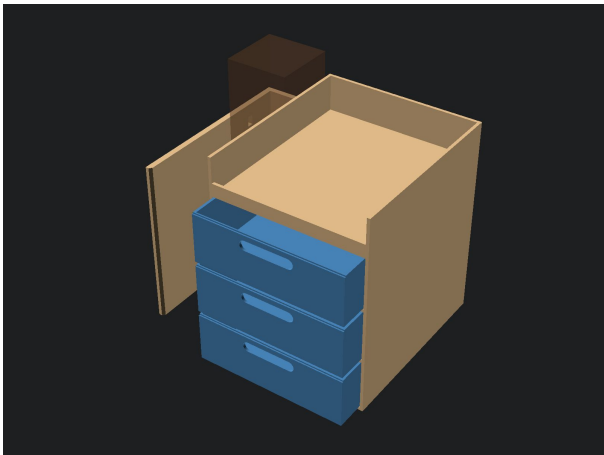
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/rect-rail-leg/housing_print.stl` and `export/rect-rail-leg/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# Deep tray top (tray-top)

Three mid bays plus a deep top tray (28 mm side/back walls, 8 mm front lip).

| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 3     | leg_on_left        | left                 |
| drawer_height mm          | 40    | leg_x mm           | 50                   |
| drawer_width mm           | 120   | leg_y mm           | 50                   |
| drawer_depth mm           | 140   | zip ties (auto)    | 3                    |
| ledge_left mm             | 28    | housing print X mm | 180.9                |
| ledge_right mm            | 28    | housing print Y mm | 157.6                |
| ledge_back mm             | 28    | housing print Z mm | 142.4                |
| ledge_front mm            | 8     | drawer print mm    | 119.1 x 139.2 x 39.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 119.1 x 139.2 x 39.5 |



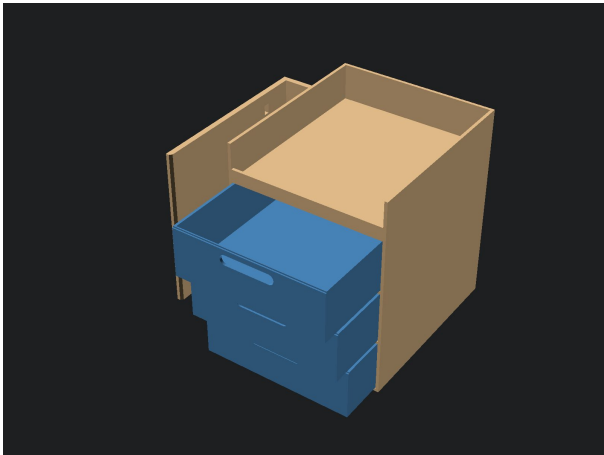
ISO, drawers pulled, ghost leg



ISO from the rear / mount

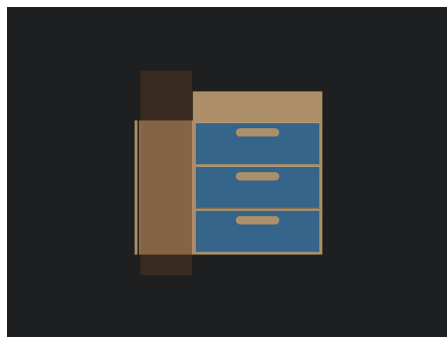


ISO opposite flank



Exploded — housing and drawers separate

## Deep tray top — drawings and print poses



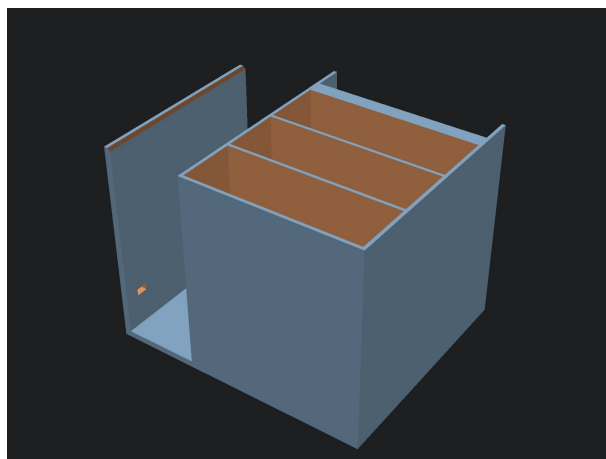
*Front elevation*



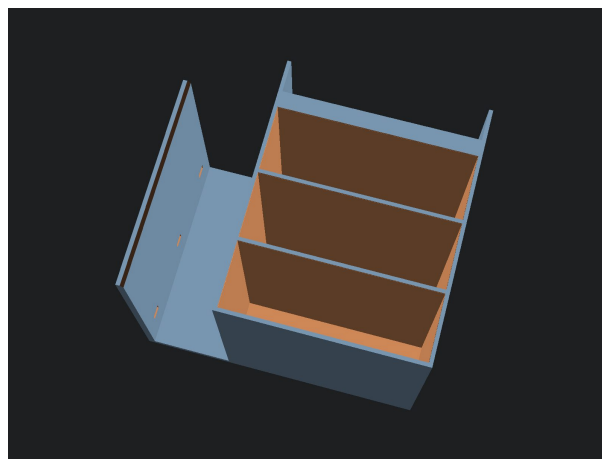
*Top / plan (C-channel)*



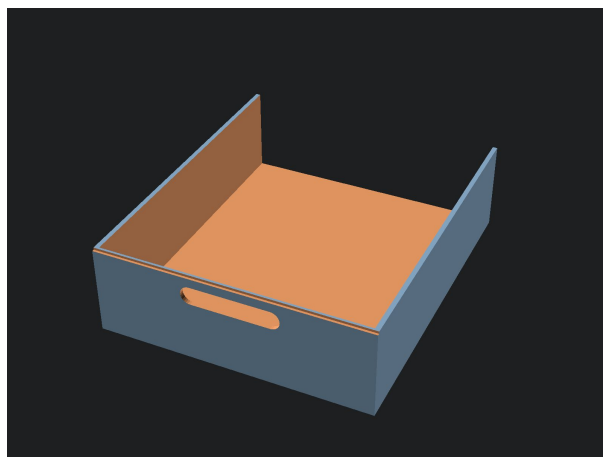
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



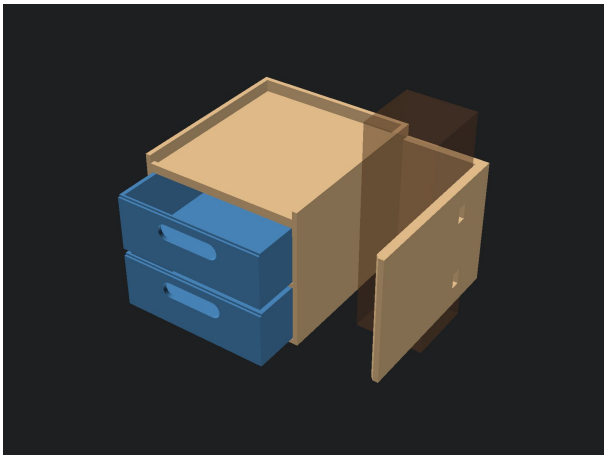
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/tray-top/housing_print.stl` and `export/tray-top/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

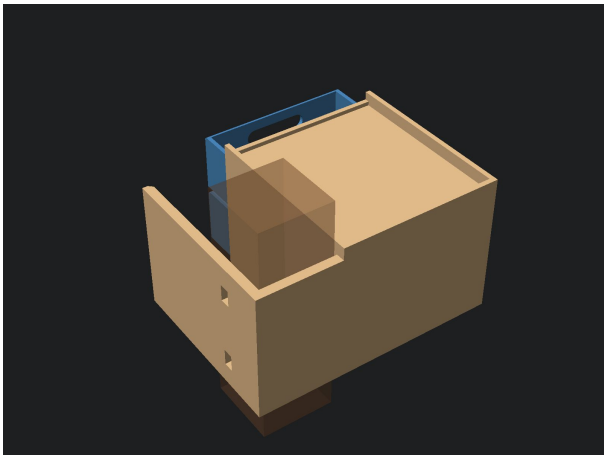
# Mini right (mini-right)

Small two-bay set for a 35 mm leg on the right. Short print, low mass.

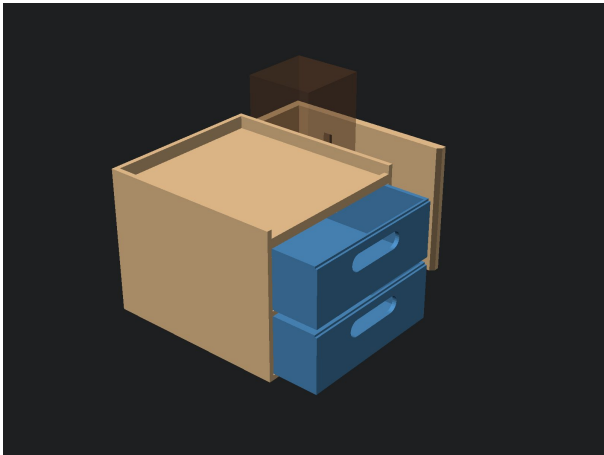
| Parameter                 | Value | Parameter          | Value              |
|---------------------------|-------|--------------------|--------------------|
| drawer_count              | 2     | leg_on_left        | right              |
| drawer_height mm          | 28    | leg_x mm           | 35                 |
| drawer_width mm           | 75    | leg_y mm           | 35                 |
| drawer_depth mm           | 85    | zip ties (auto)    | 2                  |
| ledge_left mm             | 6     | housing print X mm | 120.9              |
| ledge_right mm            | 6     | housing print Y mm | 69.2               |
| ledge_back mm             | 6     | housing print Z mm | 87.4               |
| ledge_front mm            | 2     | drawer print mm    | 74.1 x 84.2 x 27.6 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 74.1 x 84.2 x 27.6 |



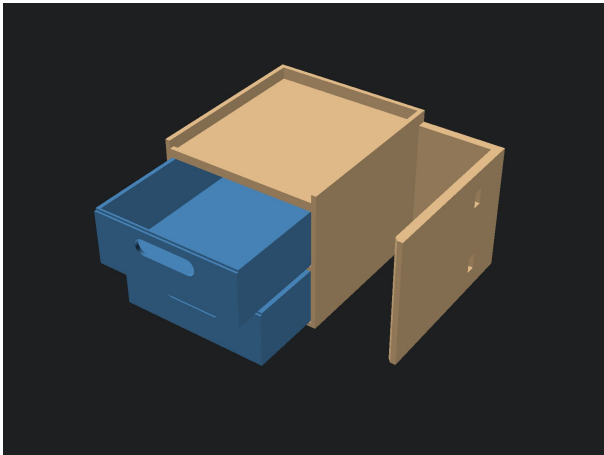
ISO, drawers pulled, ghost leg



ISO from the rear / mount



ISO opposite flank



Exploded — housing and drawers separate

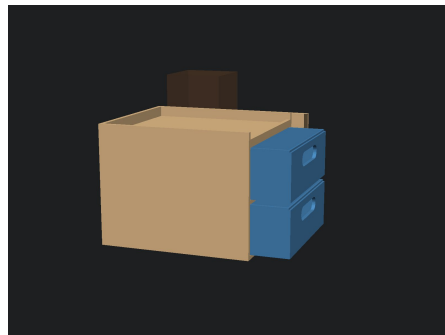
## Mini right — drawings and print poses



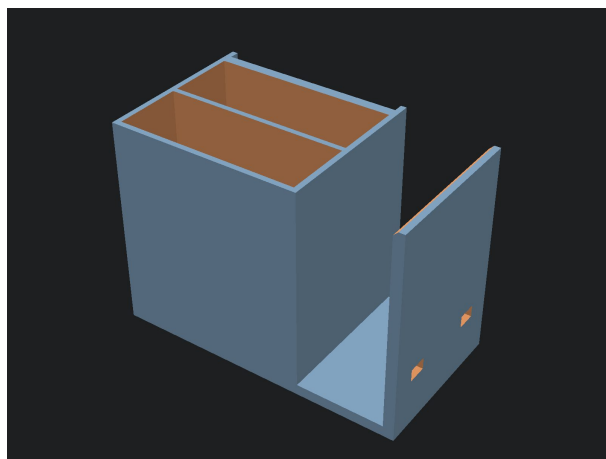
*Front elevation*



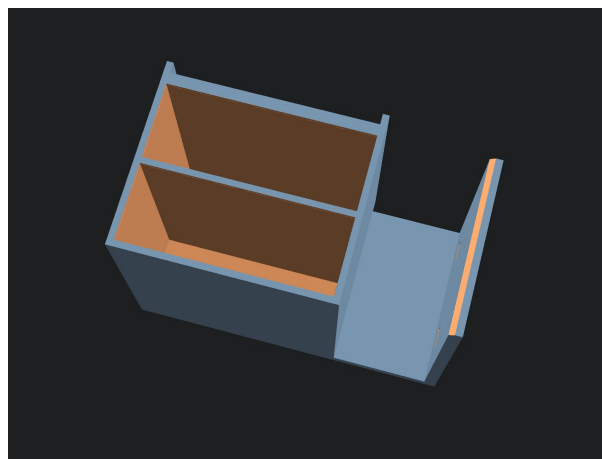
*Top / plan (C-channel)*



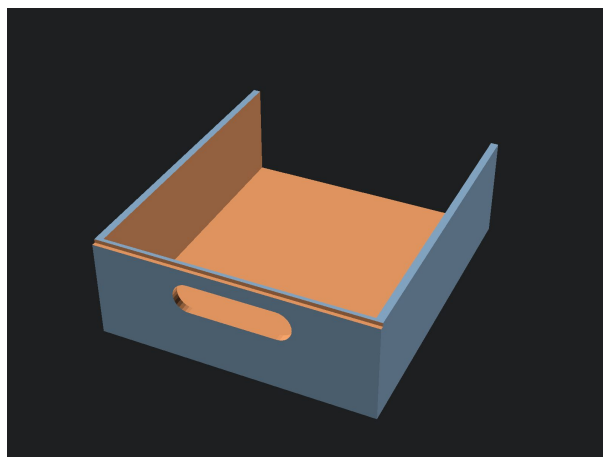
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



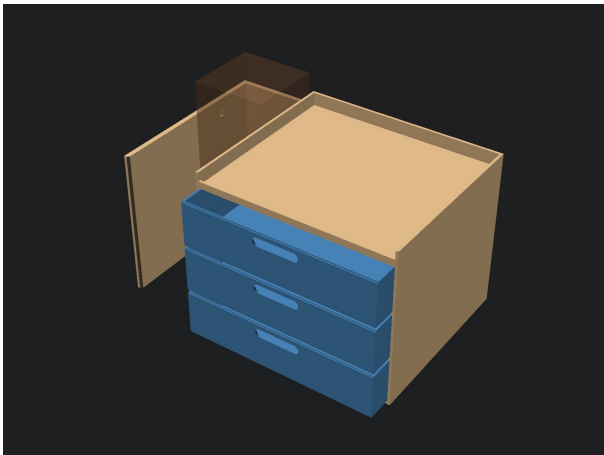
*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/mini-right/housing_print.stl` and `export/mini-right/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 2 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

# Wide three-bay (wide-three)

Wide parts bins on a 70 mm square leg. Sized to stay inside the H2D guard.

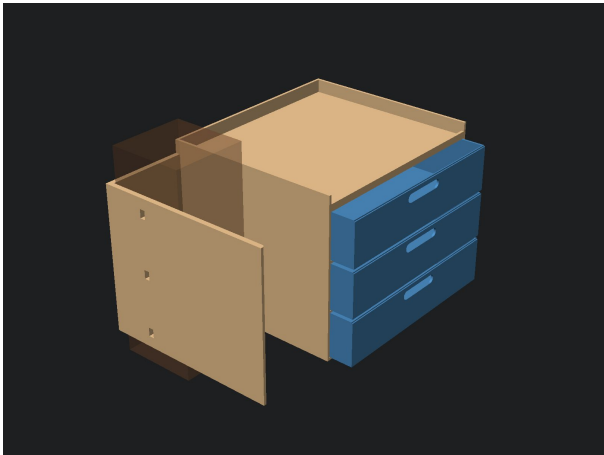
| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 3     | leg_on_left        | left                 |
| drawer_height mm          | 42    | leg_x mm           | 70                   |
| drawer_width mm           | 180   | leg_y mm           | 70                   |
| drawer_depth mm           | 160   | zip ties (auto)    | 3                    |
| ledge_left mm             | 12    | housing print X mm | 260.9                |
| ledge_right mm            | 12    | housing print Y mm | 147.6                |
| ledge_back mm             | 12    | housing print Z mm | 162.4                |
| ledge_front mm            | 4     | drawer print mm    | 179.1 x 159.2 x 41.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 179.1 x 159.2 x 41.5 |



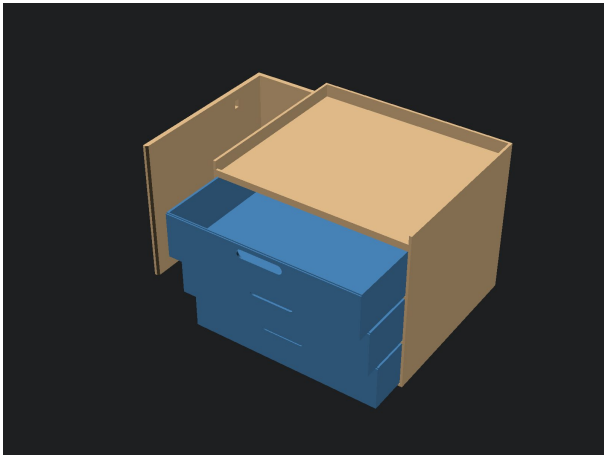
ISO, drawers pulled, ghost leg



ISO from the rear / mount

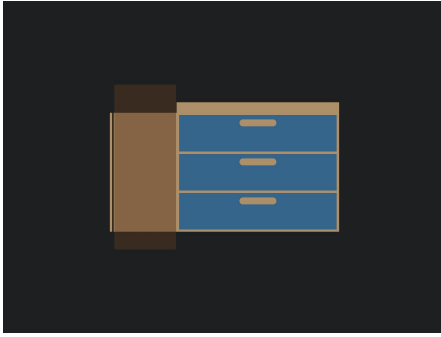


ISO opposite flank



Exploded — housing and drawers separate

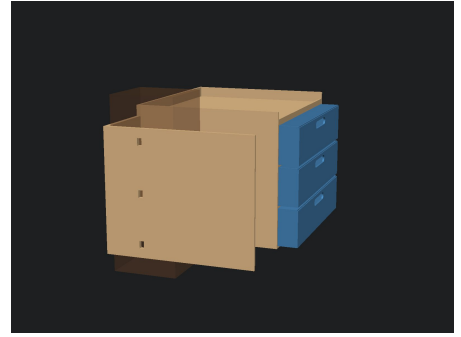
## Wide three-bay — drawings and print poses



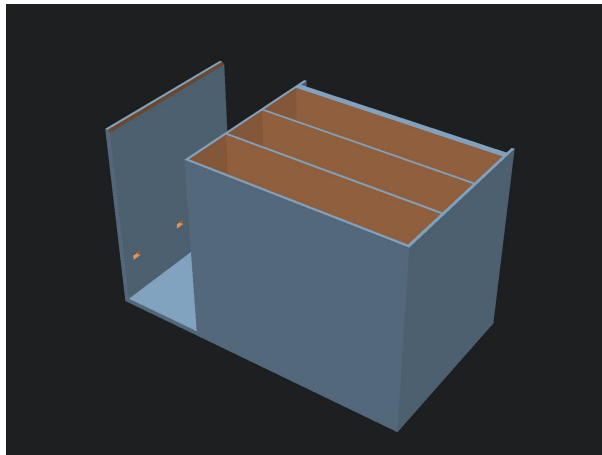
*Front elevation*



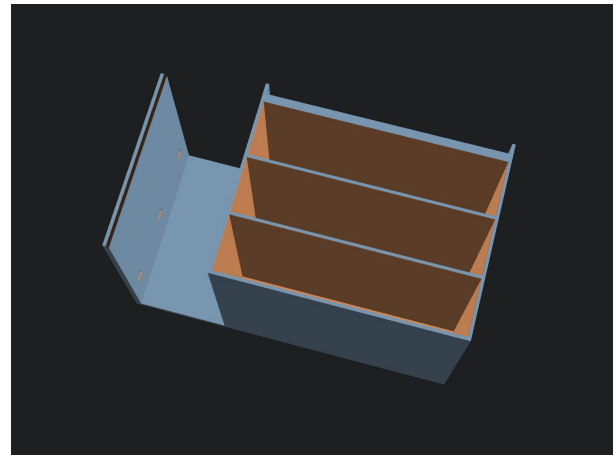
*Top / plan (C-channel)*



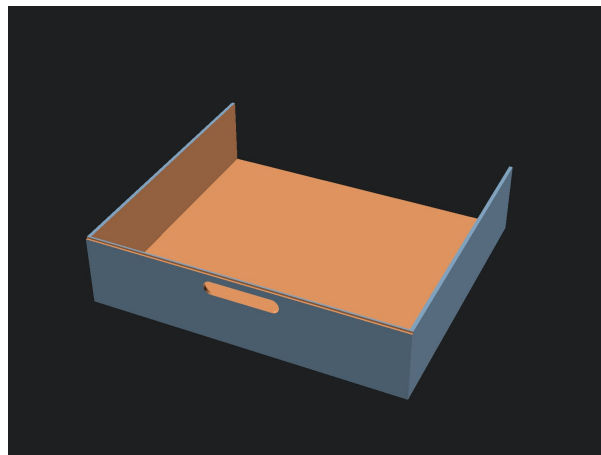
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/wide-three/housing_print.stl` and `export/wide-three/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

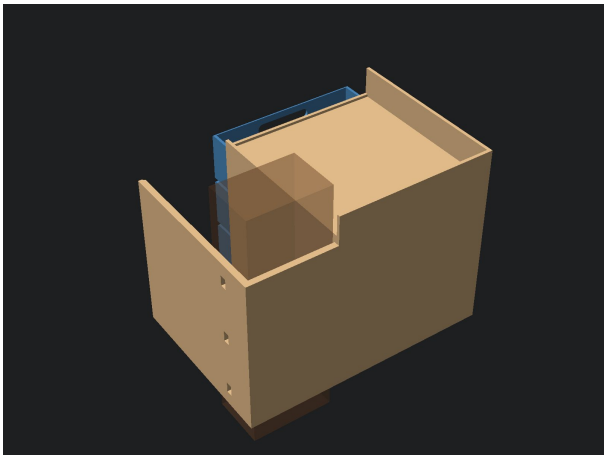
# Asymmetric wall (asymmetric-wall)

Right-hand mount, tall left and back ledges for a unit that sits against a wall.

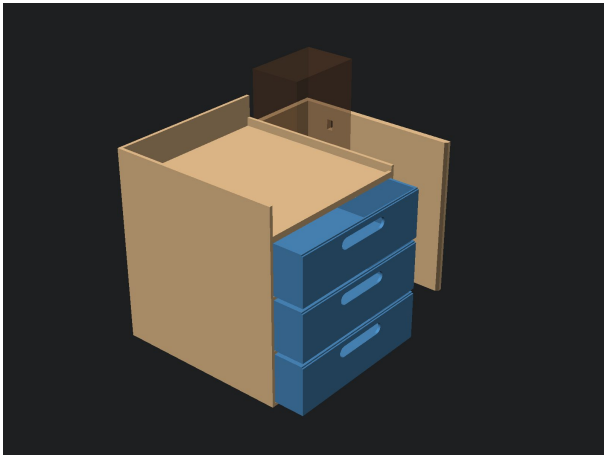
| Parameter                 | Value | Parameter          | Value                |
|---------------------------|-------|--------------------|----------------------|
| drawer_count              | 3     | leg_on_left        | right                |
| drawer_height mm          | 36    | leg_x mm           | 60                   |
| drawer_width mm           | 115   | leg_y mm           | 40                   |
| drawer_depth mm           | 125   | zip ties (auto)    | 3                    |
| ledge_left mm             | 22    | housing print X mm | 185.9                |
| ledge_right mm            | 6     | housing print Y mm | 139.6                |
| ledge_back mm             | 22    | housing print Z mm | 127.4                |
| ledge_front mm            | 2     | drawer print mm    | 114.1 x 124.2 x 35.5 |
| H2D guard 320 x 315 x 320 | PASS  | drawer outer mm    | 114.1 x 124.2 x 35.5 |



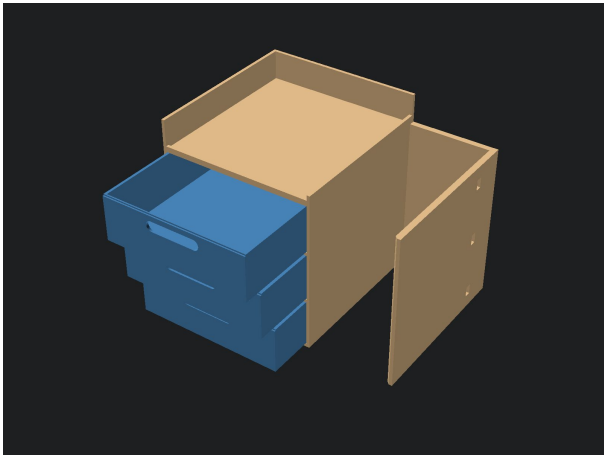
ISO, drawers pulled, ghost leg



ISO from the rear / mount



ISO opposite flank

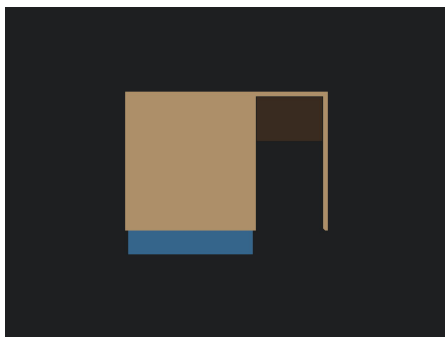


Exploded — housing and drawers separate

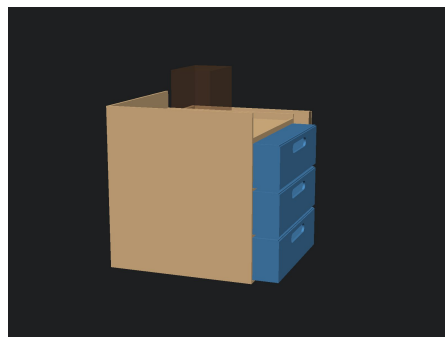
## Asymmetric wall — drawings and print poses



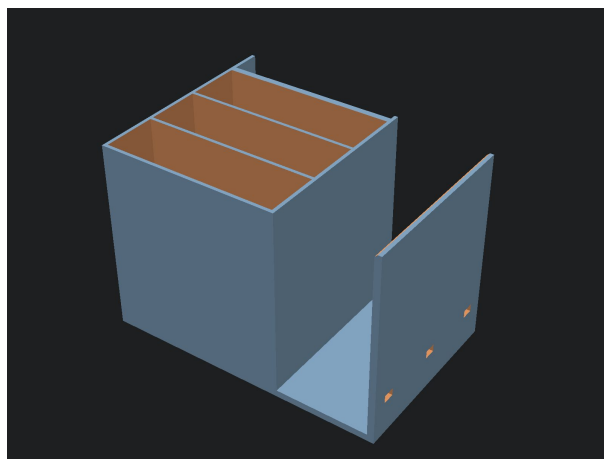
*Front elevation*



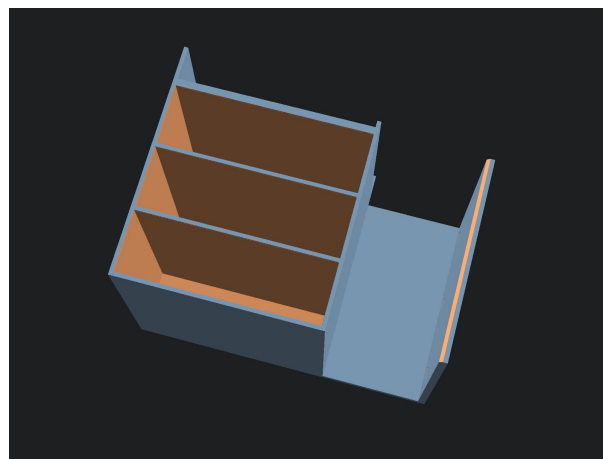
*Top / plan (C-channel)*



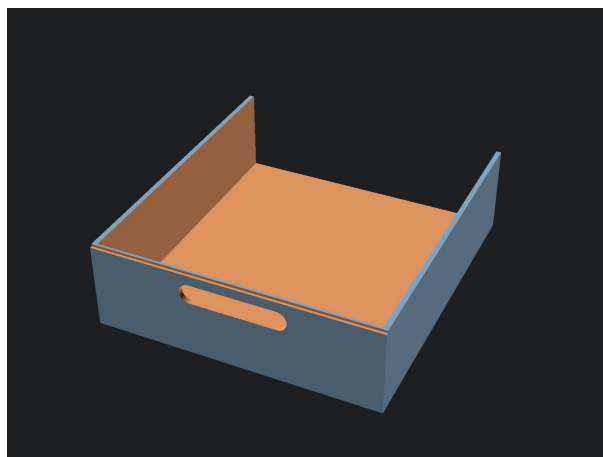
*Mount side (zip-tie windows)*



*Housing print — openings up*



*Housing print — looking into bays*



*Drawer print — bottom down, finger slot in the front wall*

Export STL: `export/asymmetric-wall/housing_print.stl` and `export/asymmetric-wall/drawer_print.stl`. Print housing as exported (openings up). Print one drawer per bay, bottom down, 3 copies. PETG recommended for the housing; PLA is fine for drawers. 0.4 mm nozzle, 0.2 mm layers, 3 walls, 20% gyroid. No supports.

## Customizer reference

Open `leg-drawers.scad` in OpenSCAD. The Customizer groups are Part, Drawers, Table leg, Top ledges, and Structure. Set `part` to `housing_print` or `drawer_print` before F6 and STL export. A variant from this catalog is a set of -D flags matching `variants.json`.

Locked print rules: housing openings face +Z after `housing_print` rotation (back of the housing on the bed, C-channel outer wall is a vertical wall, back stop is the first layers). Drawers export with the floor on Z=0. If a parameter set exceeds the H2D guard, OpenSCAD `assert` fails and no STL is written.